



ashwa RACING
LIVE IT. LOVE IT. RACE IT.

PROGRESS
IN
MOTION.

**MARCH - MAY
2026
EDITION**

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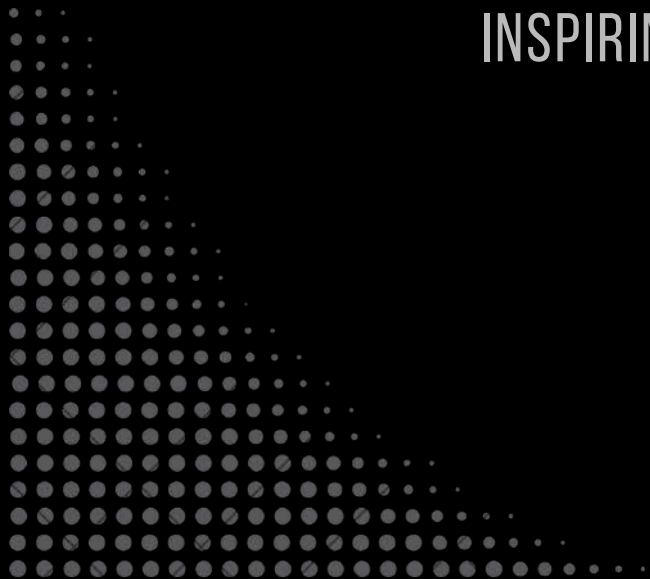


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**VISION
AND
MISSION**

GUIDING PURPOSE,
INSPIRING FUTURE.



> MISSION AND VISION <

VISION

To be the national leader in all categories of Formula Student, to push the limits of engineering and compete in events at the highest level.

To foster student mobility solutions in the future by developing sustainable prototypes across various domains.

MISSION

Prepare students for the industry by designing and building world-class vehicles while facilitating industry interaction.

Showcase India's engineering expertise in mobility on a global stage and provide talent opportunities for the industry.



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THE LEARNING CURVE

WHERE THEORY
MEETS WORKSHOP
REALITY



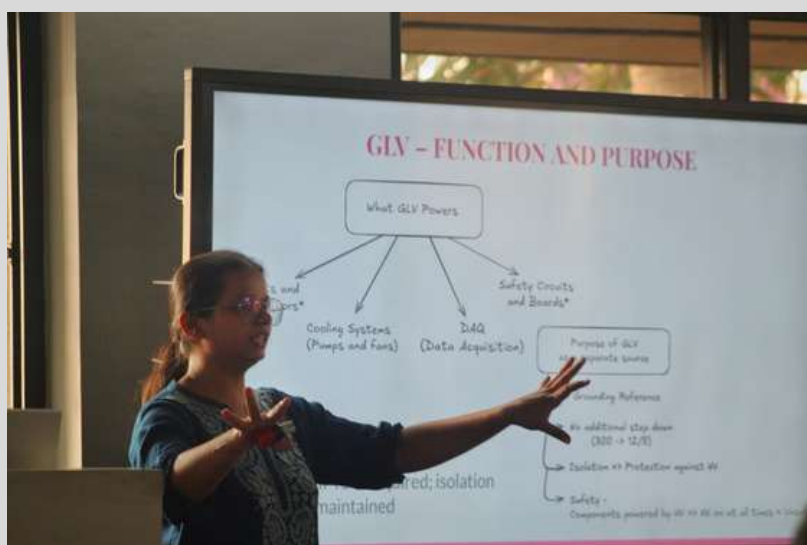


THE LEARNING CURVE BEGINS

The journey of innovation and engineering excellence continues at Ashwa Racing as we welcome a new cohort of aspiring engineers.

The Junior Training Program for this year commenced with great enthusiasm, marking the beginning of an immersive learning experience designed to shape the next generation of team members. The sessions intend to equip juniors with a thorough grasp of Ashwa Racing's vision, processes, and the teamwork ethos that propels the team ahead.

The training sessions for this year started on March 31, 2026, and concluded on May 12, 2026. Throughout this period, juniors are actively engaged in hands-on activities, interactive workshops, and practical demonstrations that bridge the gap between theoretical knowledge and real-world application. A key highlight of this year's program is the increased focus on collaborative problem-solving and peer-driven learning, encouraging participants to think critically and contribute ideas from the very beginning.



An Electrical Sub-System Engineer covering the Ground Low Voltage System.



The brakes and sensor sessions gave juniors a taste of vehicle control systems and engine monitoring. The training on brakes covered hydraulic braking systems, brake bias, master cylinders, brake lines, callipers, brake pad materials, and the working principle based on Pascal's Law. The sensors' sessions included the engine coolant temperature sensor, mass airflow sensor, and crankshaft position sensor, describing the roles and how they communicate with the ECU to keep track of the engine parameters and contribute to the smooth functioning of the vehicle



Decoding braking performance and stability.

The chassis sessions covered structural support, load transfer, rigidity, and the types of chassis used in motorsport.

Sessions on suspension covered double wishbone geometry, dampers, A-arms, and how suspension helps to maintain traction, stability, and ride control.



Interactive session on tyre force generation and vehicle dynamics.



The juniors were introduced to practical workshop knowledge with welding and workshop management sessions as well as subsystem theory. The sessions covered welding techniques including arc welding and gas welding, types of electrodes, workshop safety measures, and working with basic tools like Allen keys, spanners, mensuration tools, fasteners, and shearing tools; providing the juniors with a technical understanding and practical familiarity of the processes involved in building and maintaining a Formula Student car.

Focusing on innovation, collaboration, and technical proficiency, the training initiative lays the groundwork for juniors to develop into skilled and self-assured contributors. As the sessions advance, we anticipate seeing their development and the new viewpoints they contribute to Ashwa Racing's path.



Closing the session on a note of innovation and teamwork.

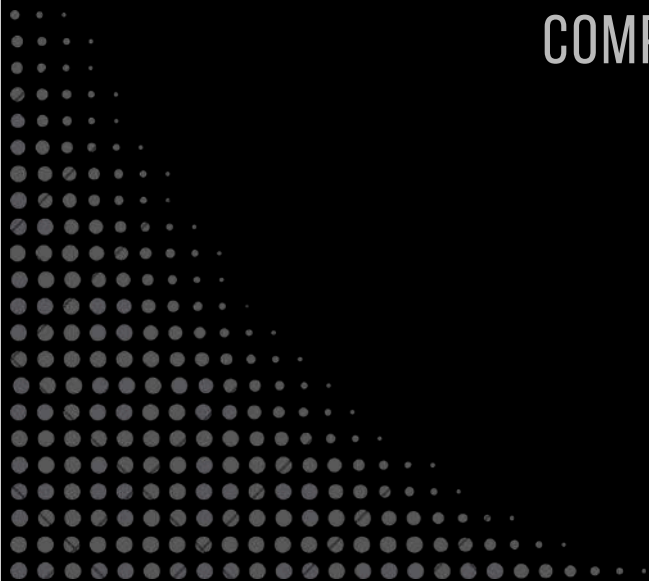


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**RZ-XX8E:
MAPPING OUT
THE MILESTONES**

FROM CONCEPT TO
COMPETITION



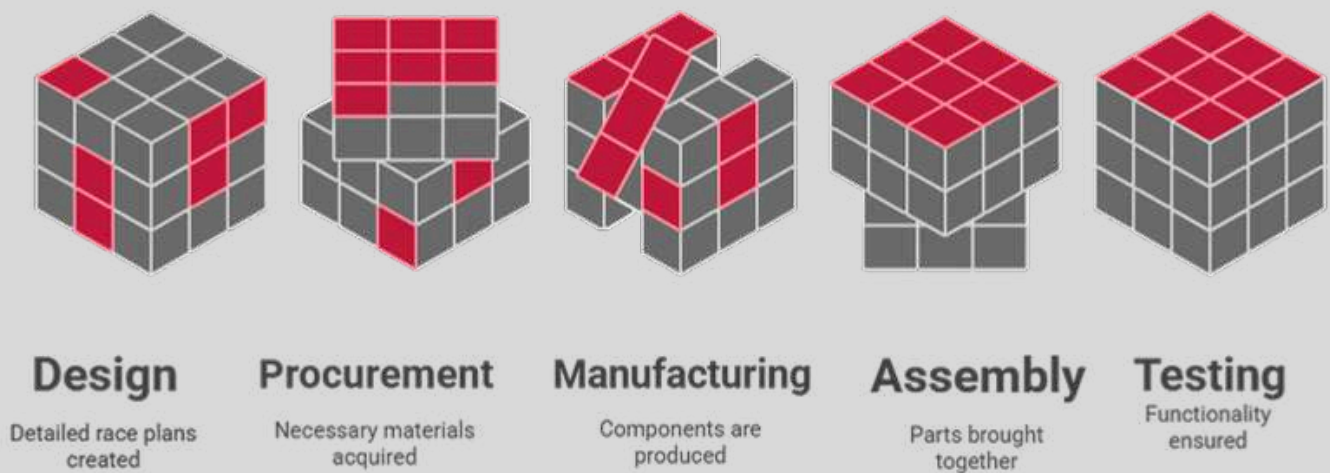


RZ-XX8E: MAPPING OUT THE MILESTONES

As Ashwa Racing prepares for the next competition season, the team has been working to establish a robust timeline to guide our journey from pre-design through to the final race day.

Team meetings were held frequently to establish clear timelines and define responsibilities, keeping everyone aligned throughout the season.

Race Roadmap



Each phase is carefully sequenced to build upon the previous one, creating a structured progression from initial concepts to race day performance. This approach ensures that technical development, resource planning, and team coordination all advance in lockstep, with built-in checkpoints to maintain quality and address challenges before they compound.

The pre-design phase serves as the foundation for the entire project. This critical phase involves:

- Assessing previous year cars through testing data and documentation.
- Setting competition goals based on a thorough analysis of past results.
- Identifying shortcomings and developing targeted solutions.
- Making architecture decisions about which components to reuse, fix, improve, or redesign.

Every decision made at this stage follows a clear principle: ideas must first be validated at the car level before breaking down into subsystem-specific goals. This ensures that all changes align with our overall team objectives.

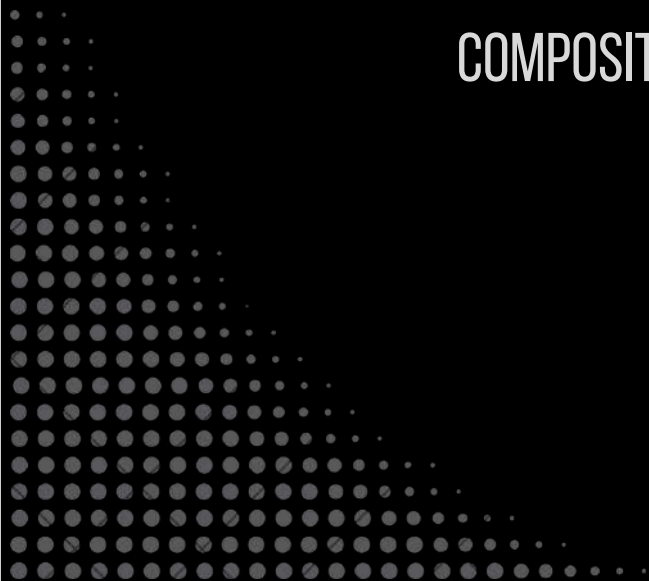


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BEHIND THE BUILD

UNDERSTANDING THE PRECISION BEHIND
COMPOSITE FABRICATION





BEHIND THE BUILD: UNDERSTANDING THE PRECISION BEHIND COMPOSITE FABRICATION

SPEEDWORKS®, our proud sponsor, is a Custom Tuning Garage based out of Bengaluru, Karnataka. They are a MRF ITC Super Stock Champion (2024 Race Season) with over 3500+ Performance Brands to cater to their customers. The highlight is the In-House AWD Dual Eddy Brake Dyno - allowing for high-torque testing with precise, rapid load control. We have been working closely with Mr Siddharth and Mr Sudeep (the co-founders, the latter of whom is our Alumni!) for the better part of this year, and they have generously shared their motorsport knowledge, providing internships for our members at their facilities. Many of our team members are actively involved with the core functions of the company, aiding in Surface Modelling and Composite Fabrication to Social Media Management and Outreach.

On 28th and 29th of March, Metacarbon, a partner organisation, hosted a two-day composite fabrication workshop at the Speedworks facility near KIA, Bengaluru. We delved into Composite categorisation, selection and working SOPs. We received practical experience on how to work with the same; up-skilling ourselves and giving us great practice that preps us for when we fabricate our body works with similar composite materials.

Day 1: Getting Up-to Speed

The first day of the workshop was dedicated to foundational techniques. We began by organising our workspace and gathering the specialised tools and materials required for the carbon fiber fabrication process. The first step in any carbon fiber project is mould preparation. To understand the efficacy of different release agents, we performed a comparative test on the front hood of a Toyota Fortuner.



A car bonnet with an intake opening (hood scoop)



After a thorough cleaning with soap and water to remove surface contaminants, we divided the hood into two sections:

- Wax application: On the first half, we applied automotive-grade polish wax. This required methodical, circular scrubbing to ensure a consistent, non-stick barrier.
- Polyvinyl Alcohol (PVA): On the second half, we applied PVA using a brush. This technique required gentle, linear strokes to achieve a uniform film while preventing air bubbles from forming.

This comparison was vital for our team to evaluate which medium offered the best balance of ease-of-use and release reliability for our specific workshop applications. We shifted focus to the actual production mould for the Toyota Fortuner hood. While the mould was already cast, it required significant surface refinement.

The mould was washed and dried to provide a clean baseline. Using high-grit sandpaper to “wetsand” the mould. We paid particular attention to the corners and radii where surface roughness was most prominent. Once the mould surface was perfectly smooth, we applied the final layer of PVA, maintaining strict linear strokes to ensure a flawless, bubble-free interface.

By the end of the day, the mould was dried, inspected, and primed for the layup process. With the preparation complete, the team concluded Day 1, ready to begin the carbon fiber lamination the following morning.



Application of the the PVA film on the model surface

Day 2: In the Power-band



With fresh, new minds joining the workshop on the second day, Mr Sudeep gave a quick run-through of the previous session and set the pace for what was coming up next. He explained surface preparation, how this must be done in the utmost and sincere manner, because a bad mould means an even worse part. First and foremost, it should be thoroughly cleaned with the help of IPA (Isopropyl Alcohol). Even the tiniest dust particles can leave a significant gap within the mould compromising the end product quality.

PVA is the most commonly used release agent, which tends to leave a thin layer of film that easily peels off - the issue lies in getting the perfect ratio to mix with water (the dilution controls the thickness, i.e., viscosity of the solution) as well as the time you have to apply it. Using a brush applicator with large, slow, purposeful strokes is good practice; short, fast, random strokes introduce air bubbles, which lead to poor mould quality. Wax is an alternative that was also explored. Then we covered mould dimensions; length dimensioning precedes width dimensioning, Infusion Sheets and how we can use ubiquitous mosquito nets as a cost-effective alternative that achieves the same result. The Peel Ply concept was introduced; a subtle but very important observation was made that the resin absorption can be tracked using the dotted lines provided on the sheet.



Peel Ply is meticulously arranged and fit to account for even the most minute surface topological features

It was explained to us that a mesh surface acts as a barrier between the top layer and peel ply layer, which does not deform in the vacuum.



Sealant Tape (Butyl Base Rubber Tape) is used as an adhesive in this process and is often two-sided. There must be a few gaps present, just enough to accommodate the Infusion Pipes for the Resin.

The resin under consideration is UV-sensitive, which is very important because this material naturally yellows. Resin formation is an exothermic reaction and hence cures faster in heat.

Everyone was eagerly waiting for the Carbon Fibre to come out, and when it did, we were all quick to spread the sheet out and immediately get to work on it.



Masking tape at the edges of the carbon fibre sheet to prevent fraying



Mr Sudeep explained that there are several grades that are classified based on their GSM and weave patterns.

He gave several precautionary instructions, such as wearing protective equipment (masks and gloves), how to be careful not to step or put pressure on the fibre, the correct method of cutting the fibre by applying masking tape and cutting across the middle to avoid fraying of fibre strands.

It made us realise it is often the smallest and simplest of instructions that, if followed with utmost discipline, lead to the best results.



Carbon Fibre Sheet is laid down atop a plastic wrap to prevent dust from contaminating the surface

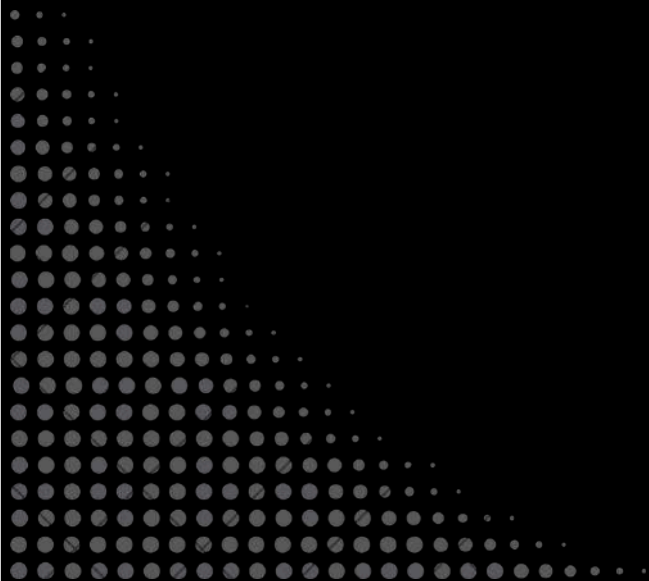


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MASTERS AT THE DERBY

TRACKMEISTERS - PERFECT LAP SESSION 2





MASTERS AT THE DERBY TRACKMEISTERS - PERFECT LAP SESSION 2

Ashwa Racing was invited out to GP1 Performance Zone, Bengaluru on Sunday, 17th of May for the Perfect Lap Session 2 organised by Trackmeisters. It was a time-attack event featuring custom built race cars from renowned garages (such as Speedworks, Race Concepts, mspeedPerformance, Code6 Racing) driven by racing drivers. Automotive Enthusiasts were welcome to register their own vehicles and get a first hand experience of what it means to push a car to its absolute limits. Visitor registrations were also open to all, free of charge.

It was chilly Sunday morning at The Workshop when we loaded the car for transport. Immediately after reaching the venue we got the car ready and set-up under a comforting shade for its static display.

Crowds started building up slowly but steadily around the car. Ex-formula student engineers, current motorsport racers, and enthusiast parents with curious children; everyone was suprised to see a Formula Student Car among the mix of production vehicles at the event.



Team-members were quick to interact and discuss with the audience; taking advices from people in the industry, answering inquisitive questions by explaining complex engineering concepts in simple words.

We were also interviewed by fellow content creators, who helped convey our story to a wider audience.

This is how Ashwa Racing chooses to build its network, **interacting and communicating with people.**

The day continued with team members enjoying the competitive racing that was showcased by the participants not to mention the plethora of distinct modifications that the custom-tuned cars were running.

It was interesting to see how each driver decided to take their out-laps and hot-laps, it brought back memories from the many national and international we've competed at.



As it was time to roll the car back out to the truck for transport back to the workshop, the team felt a wave of gratitude and contentment. We left the event with good memories, great advice and new connections.



**Ashwa Racing sincerely thanks Trackmeisters for this invitation
We look forward to further collaboration**



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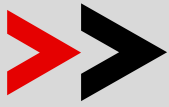
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**TEAM
DIRECTORY
2026**

DRIVEN BY PASSION, UNITED BY ENGINEERING—
THIS IS ASHWA RACING!

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